

RAGHUMANI MEHTA

Vancouver, BC

☎ +1 (236) 863-4315 ✉ raghumani2003@outlook.com [in linkedin.com/in/mehta-raghu](https://www.linkedin.com/in/mehta-raghu) github.com/raghumanimehta
🌐 m-raghumani.vercel.app

Education

University of British Columbia

B.Sc. in Computer Science, Minor in Physics

- Dean's List (2024–2025)

Expected May 2028

Vancouver, BC

Technical Skills

Languages: C, C++, Java, Python, TypeScript, JavaScript, SQL, Rust

Frameworks & Tools: Node.js, Express, Spring Boot, Git, Docker, ROS, GitHub Actions, FreeRTOS, Pico SDK

Systems: Linux/Unix, POSIX, Sockets, Real-Time Scheduling

Cloud: AWS (Textract, Lambda, S3, SQS, SNS, RDS)

Testing: JUnit, Mocha/Chai, Pytest, GDB, Valgrind, Raspberry Pi Debug Probe

Experience

Systopia Lab, UBC

May 2026 – Present

Undergraduate Research Assistant

Vancouver, BC

- Contributing to systems research on protecting confidential cloud applications from memory access pattern leakage.
- Exploring user-level paging as a practical way to hide memory access patterns without requiring major application rewrites.
- Preparing to prototype paging-based obfuscation techniques and evaluate their performance overhead on representative cloud workloads.

UBC Sailbot

Feb. 2025 – Present

Software Lead

Vancouver, BC

- Lead a 30-member software team across 5 subteams, coordinating development, code reviews, and integration planning for an autonomous sailboat.
- Design autonomy software architecture and align software interfaces with electrical and mechanical subsystem constraints.
- Guide development of route-planning algorithms for autonomous sailing under uncertain wind and marine conditions.
- Built mock testing and simulation workflows to validate navigation logic before hardware integration.

UBC Department of Computer Science

Sept. 2025 – Apr. 2026

Undergraduate Teaching Assistant

Vancouver, BC

- CPSC 317: Computer Networking (Jan. 2026 – Apr. 2026): Led tutorials and supported students on IP forwarding, UDP, TCP, sockets, encryption, client-server programming, and debugging C/Java networking assignments.
- CPSC 310: Software Engineering (Sept. 2024 – Dec. 2025): Mentored 50+ students on TypeScript, Node.js, testing, and CI workflows across a large software engineering project.

Projects

FreeRTOS Scheduler Extensions | *C, FreeRTOS, Real-Time Systems, Raspberry Pi Pico H* **Feb. 2026 – Apr. 2026**

- Extended the FreeRTOS kernel with EDF, SRP, CBS, Global EDF, and Partitioned EDF scheduling policies.
- Implemented runtime stack sharing for SRP to reduce stack overhead while preserving resource-access correctness.
- Enabled and tested multicore scheduling behavior for both Global EDF and Partitioned EDF on Raspberry Pi Pico H.
- Thoroughly tested scheduler behavior using task timelines, resource-access traces, and low-level debugging with the Raspberry Pi Debug Probe.
- Repository: github.com/alberto-escobar/CPSC538G

Reliable UDP | *C++, Linux/Unix Sockets, POSIX*

Jun. 2025 – Present

- Building a reliable file-transfer layer over UDP with acknowledgments, sequencing, retransmissions, and out-of-order packet handling.
- Implementing custom sender and receiver components using POSIX sockets to provide TCP-like delivery guarantees.
- Testing packet loss, packet reordering, and dropped acknowledgments to validate correctness under unstable network conditions.
- Repository: github.com/raghumanimehta/ReliableUDP